Alexander Poulin

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Education

Northeastern University – *Khoury College of Computer Science (Honors Program)* May 2027
Candidate for Bachelor's in Computer Science & Physics, Minor in Astrophysics Boston, MA

Relevant Coursework*:* Object-Oriented Design, Algorithms, Calculus III, Diff-Eq. & Lin-Alg., Prob. & Stats., Thermo. & Stat. Dynamics, Quantum Mech., Electronics, Artificial Intelligence, Astrophysical Processes, Radio Astronomy, Multi-Messenger, Advanced Quantum Mech., E&M, General Relativity & Cosmology.

Research Experience

DAWN-IRES Scholar Jun – August 2026
Cosmic DAWN Center / Technical University of Denmark (Advisor: Shuowen Jin & Marko Shuntov) Copenhagen, DK
Studied the nature of unknown, dark, and red objects in COSMOS-Web, selected via a F277W dropout. Utilized multi-wavelength, sub-mm data to constrain SED and stacked SED fits.

Determined 2/29 of the selected objects to be DSFG at a redshift z~5. The remaining 27 objects were point-like sources where 8 exhibit stacked photometry that suggest LRD z>7.

Undergraduate Research Co-op Jan – Present
Boston University Cosmology (Advisor: Professor Tereasa Brainerd) Boston, MA
Investigating the spatial distribution of satellite galaxies as a function of redshifts in the galaxy clusters of the TNG300 and TNG-Cluster simulations to determine its feasibility as a tracer for dark matter and the origins of its lopsidedness.

AJC Merit Scholarship Co-op Jan – Jun 2025
CERN/MIT/Northeastern (Advisor: Professor Skinnari for CMS) Boston, MA
Improved signal-to-background separation in FCC's IDEA detector with preliminary results indicating a 70% decrease in background hits, by classifying beam backgrounds and designing a spatial recognition algorithm.

Undergraduate Research Oct 2024 – Apr 2025
Jacqueline McCleary Group (Advisor: Professor Bryanne McDonough) Boston, MA
Advanced understanding of galaxy evolution, shown through differentiated magnetic field profiles of satellite vs. central galaxies, by analyzing TNG100-1 simulations pre- and post-infall.

Undergraduate Research Nov 2023 – Present
DeLTA Laboratory (Advisor: Professor Alberto De La Torre Duran) Burlington, MA
Developed a graphical interface data analysis toolkit, CLADA, for Cornell's CHESS beamline with nxs-analysis-tools, focusing on a clean workflow towards delta-pdf capabilities.

Investigated substrate-dependent ARPES behavior in MnTe, validated by contrasting magnetic responses on SrF_2 vs. GaAs, by conducting comparative ARPES experiments.

Increased efficiency of ARPES data workflows, measured by streamlined processing and visualization, by developing ARDA, a Python-based analysis toolkit for NanoESCA data.

Honors and Awards

DAWN-IRES Scholars Program May – Aug 2026
Barry Goldwater Nomination Jan 2026
Four Gavin Brothers Experiential Learning Scholarship Jan 2026
John Martinson Honors Program, Northeastern University Nov 2024 – Present
PEAK Award Summit Sep 2025 – Present
AJC Merit Research Scholarship Jan – June 2025
PEAK Award Ascent Sep – Dec 2024
PEAK Award Base Camp May – Aug 2024

Presentations

Cosmic DAWN Center, Copenhagen, Denmark, August 2026. **Poulin, Alexander,** Shuowen Jin, Marko Shuntov. "A Multi-Wavelength Study of Extremely Red and Dark Objects in COSMOS-Web" (Cake-Talk).

DAWN Summit, Copenhagen, Denmark, August 2026. **Poulin, Alexander,** Shuowen Jin, Marko Shuntov. "A Multi-Wavelength Study of Extremely Red and Dark Objects in COSMOS-Web" (Flash-Talk).

Northeastern University RISE, Boston, MA, April 2024. **Poulin, Alexander**. "Investigating Enhancements to the Tracking System of the FCC and CMS Detectors" (Poster).

Northeastern University RISE, Boston, MA, April 2024. **Poulin, Alexander**. "Determining the Role of the Substrate in the Electronic Structure of MnTe" (Online Poster).

Northeastern University RISE, Boston, MA, April 2024. Sununu, August, **Poulin, Alexander**, Cherry, Madison. "Design and Development of a Cost-Effective Electrocardiogram Simulator Using Arduino Uno" (Poster).

American Physical Society Northeast, Boston, MA, October 2024. **Poulin, Alexander**. "Deploying the NanoESCA Spectrometer to the Novel Altermagnet MnTe" (Poster).

Publications

Alexander Poulin, Shuowen Jin, Marko Shuntov. (In Preparation). "A Multi-Wavelength Study on F277W Dropouts in COSMOS-Web." Target Journal: Astrophysical Journal

Alexander Poulin and Tereasa G. Brainerd. (In Preparation). "Lopsided Satellite Distributions Across Redshift in TNG300 and TNG-Cluster simulations." Target Journal: Astrophysical Journal

Sara Bey, Shelby S. Fields, Nicholas G. Combs, Bence G. Márkus, Jiashu Wang, Liam Schmidt, Lincoln Curtis, Allecia Dodd-Noble, **Alexander Poulin**, Syed Mohammad Shahed, Resham Regmi, Mariia Holub, Phillipe Ohresser, Arun Bansil, Swastik Kar, Haile Ambaye, Valeria Lauter, László Forró, Cory D. Cress, Joseph C. Prestigiacomo, Nirmal Ghimire, Alberto de la Torre, Steven P. Bennett, Xinyu Liu, Badih A. Assaf. "Large anomalous Hall effect in α-MnTe thin films grown on SrF_2(111)". Phys. Rev. B 114, 194403 (2026).

Bryanne McDonough and **Alexander Poulin**. "Magnetic Fields of Satellite Galaxies Stronger than Comparable Centrals in TNG100." 2025 Res. Notes AAS 9 93

Leadership

AeroNU – Liquid Rocket Engine Software Team Lead

Organized two weekly meetings focused on teaching flight and engine software and completing hands-on projects. Developing an engine and flight control software for a liquid rocket engine using C++ to control custom PID needle valve, monitor pressures systems, and send commands.

Engineering Lead – NASA L'Space Program

Led the engineering team and our CAD design of a lunar dust tolerant connection to withstand lunar regolith improving durability and performance with a magnetic twist connection and a co-written proposal.

Kiddo Byte – Instructor

Co-instructed three classes of ~15 students aged 8-15 years old from disadvantaged schools in the Boston area. Instructed basic computer science principles and coding through the Scratch. Additionally helped raise $14,000 to support funding computers and computer accessories for students in Boston, Kenya, and Ukraine.

Organizations

Clubs: AeroNU, Society of Physics Students (SPS), Signing Huskies, Student's Exploration and Development of Space (SEDS), STEMout, NUSci

Technical Skills

Computational: Python, Java, MATLAB, HTML, JSS, C++, SQL, Arduino, C, PyTorch, Julia, ROOT, ROS

Applications: Git, Linux, CIGALE, EAZY-py, DS9, TopCAT, CARTA, GNU Radio, Fusion360, Blender, PyQt6, Docker, Android SDK, Siemens NX, LabView